ROBOARM — FULL WIRING (REV D)

Every board pin shown. Wires drawn between parts. Power rails use KiCad power symbols (+24V, +5V, +6V, GND). Unused pins carry no-connect flags.

UART bus A: Mega Serial2 (TX2 D16 via 1k, RX2 D17) -> PDN/UART of DRV1–3 (addr 0..2).
UART bus B: Mega Serial3 (TX3 D14 via 1k, RX3 D15) -> DRV4–5. Gives per-joint load (StallGuard), temperature and open-wire flags.

E-STOP: pressing SW1 opens the relay coil circuit, K1 drops out
and the red 24 V bus goes dead. PI and Mega stay on.
A broken E-stop wire also stops the motors (fail-safe).

PS1
+V
-V
GND
Lab bench supply: 24 V, 3 A limit (later: Mean Well LRS-200-24)
F0
10 A main
K1
30_COM
85_coil+
87_NO
86_coil-
24 V 30 A relay
+24V
GND
SW1
E-STOP (NC, latching)
PS1 for first tests: existing bench supply, 24 V, current limit 3 A.
Its output on/off button cuts motor power only; PI and Mega stay on.
The 24 V goes ONLY to the motor drivers and the servo buck. The PI has its own USB-C supply (PS2).
D0
1N4007
GND
24 V BUS (red): bench supply + -> F0 -> VM on every driver. Ground return (black): every driver GND -> bench supply -

PS2
5.1V
GND
Official PI 27 W USB-C supply

U1
USB-C_5V_IN
USB-C_GND
1_3V3
3_GPIO2_SDA
5_GPIO3_SCL
7_GPIO4
9_GND
11_GPIO17
13_GPIO27
15_GPIO22
17_3V3
19_GPIO10_MOSI
21_GPIO9_MISO
23_GPIO11_SCLK
25_GND
27_ID_SD
29_GPIO5
31_GPIO6
33_GPIO13
35_GPIO19
37_GPIO26
39_GND
USB-A_VBUS
USB-A_D+
USB-A_D-
USB-A_GND
2_5V
4_5V
6_GND
8_GPIO14_TXD
10_GPIO15_RXD
12_GPIO18
14_GND
16_GPIO23
18_GPIO24
20_GND
22_GPIO25
24_GPIO8_CE0
26_GPIO7_CE1
28_ID_SC
30_GND
32_GPIO12
34_GND
36_GPIO16
38_GPIO20
40_GPIO21
Raspberry PI 5 8GB

PI USB-A to Mega USB-B cable:
powers the Mega and carries commands.

U2
IOREF
RESET
3.3V
5V
GND
GND
VIN
AREF
GND
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
A13
A14
A15
D0/RX0
D1/TX0
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
D14/TX3
D15/RX3
D16/TX2
D17/RX2
D18/TX1
D19/RX1
D20/SDA
D21/SCL
D22
D23
D24
D25
D26
D27
D28
D29
D30
D31
D32
D33
D34
D35
D36
D37
D38
D39
D40
D41
D42
D43
D44
D45
D46
D47
D48
D49
D50/MISO
D51/MOSI
D52/SCK
D53/SS
5V
5V
GND
GND
Arduino Mega 2560 R3

DRV1
DRV2
DRV3
DRV4
DRV5
EN
MS1
MS2
PDN/UART_RX
PDN/UART_TX
CLK
STEP
DIR
VM
GND
B2
B1
A1
A2
VDD/VIO
TMC2209 module (StepStick)

R21 10k
R22 10k
R23 10k
R24 10k
R25 10k
R31 1k (UART-TX)
R32 1k (UART-TX)
+5V
GND

C20 100uF 35V
C21 100uF 35V
C22 100uF 35V
C23 100uF 35V
C24 100uF 35V

M1
M2
M3
M4
M5
B-_(coil_B)
B+_(coil_B)
A+_(coil_A)
A-_(coil_A)
17HS19-1684S-PG5
J1 Base: 0.52 Nm 1.68 A × 5.1Ω:1, set 1.4 A RMS
17HS19-1684S-PG14
J2 Shoulder: 0.52 Nm 1.68 A × 13.73:1, set 1.4 A RMS
J3 Elbow: 0.52 Nm 1.68 A × 5.1Ω:1, set 1.4 A RMS
17HS08-1004S
J4 Wrist: 0.16 Nm 1 A, set 1.0 A RMS
11HS12-0674S
J5 Wrist twist: 0.06 Nm 0.67 A, set 0.67 A RMS

Joint encoders: AS5600 reads a magnet on each joint OUTPUT shaft (after the gearbox), so the angle is absolute at power-up.
OUT = analog angle (0–5 V) to Mega A0–A4 through 1k + 100 nF at the Mega end. DIR tied to GND sets direction.
Check the module runs at 5 V (VCC) before buying; SDA/SCL left free for later I2C use.

ENC1
ENC2
ENC3
ENC4
ENC5
OUT
SDA
SCL
GPO
PGO
VCC
GND
DIR
J1 encoder, AS5600 module
J2 encoder, AS5600 module
J3 encoder, AS5600 module
J4 encoder, AS5600 module
J5 encoder, AS5600 module

R1 1k
R2 1k
R3 1k
R4 1k
R5 1k
C1 100nF
C2 100nF
C3 100nF
C4 100nF
C5 100nF
+5V
GND

FAN1
40 mm 5 V fan over driver row

U6
IN+
IN-
OUT+
OUT-
XL4015 buck, set 6.0 V
F1 2 A
F2 3 A
+6V
C7 1000uF
C8 100nF
R13 1k
J6
PWM
6V
SDA
SCL
Tool plug, GX12 6-pin
MG996R gripper servo
Gripper power: 24 V bus -> F1 -> U6 buck (set 6.0 V with a meter BEFORE connecting the servo) -> F2 -> 6 V (orange) -> C7, C8, servo.

BILL OF MATERIALS (electrical parts; full list, links and prices in hardware/bom)

Ref	Qty	Part	Note
U1	1	Raspberry PI 5 8GB	owned
PS2	1	Official Raspberry PI 27 W USB-C supply	owned
U2	1	Arduino Mega 2560 R3	owned
PS1	1	24 V bench supply	owned
DRV1–DRV5	5	BIGTREETECH TMC2209 V1.3 + heatsink	
M1, M3	2	STEPPERONLINE 17HS19-1684S-PG5	J1, J3
M2	1	STEPPERONLINE 17HS19-1684S-PG14	J2
M4	1	STEPPERONLINE 17HS08-1004S	J4
M5	1	STEPPERONLINE 11HS12-0674S	J5
W7	1	MG996R servo	J6 gripper tool
J6	1	GX12 6-pin tool plug	
U6	1	XL4015 5 A buck, set 6.0 V	
ENC1–ENC5	5	AS5600 magnetic encoder module + diametric magnet	
F0–F2	3	Inline ATO fuse holders, 10 A, 2 A and 3 A fuses	
SW1	1	E-stop, 22 mm latching mushroom, 1 NC	
K1	1	24 V coil relay, normally open, with socket	
D1	1	1N4007 flyback diode	
C7, C20–C24	6	Electrolytic 1000 uF 16 V (x1), 100 uF 35 V (x5)	
C1–C5, C8	6	Ceramic 100 nF	
R1–R5, R13, R21–R25, R31–R32	13	1k (x8), 10k (x5) metal film 1/4 W	
FAN1	1	40 mm 5 V fan over driver row	

NOTES
1. Mega pins: STEP D22–26, DIR D30–34, EN D36–40 (LOW = driver on; 10k pull-ups keep drivers OFF while the Mega boots). encoders A0–A4, gripper PWM D9, tool I2C D20–D21.
2. TMC2209 standalone mode: UART mode: MS1/MS2 set each driver bus address (0–2). microstep is set over UART; CLK = GND selects the internal clock.
3. Check pin names against the silkscreen of the TMC2209 modules you buy (vendors differ). Set phase current with the module trimmer before connecting the arm.
4. TMC2209 VM max is about 29 V; 24 V only. Never plug or unplug a motor with 24 V present. Verify coil pairs with an ohmmeter.
5. Modules sit in female headers on perfboard. Heatsink on every module. 40 mm fan over the row.
6. All grounds common. Never connect 24 V or 6 V to the Mega 5 V pins.

Module wiring — not a validated machine
RoboArm
Sheet: /
File: custom_arm.kicad_sch
Title: RoboArm — full wiring
Size: A1 | Date: 2026-11-02 | Rev: D
KiCad E.D.A. 9.0.9 | Id: 1/1